IIT JODHPUR

Major Examination: EEL71090 ML in VLSI CAD (Apr'26)

Guidelines (Total time: 3 Hours, Maximum Marks: 50):

- Please read the question paper very carefully. All questions carry equal marks. For each question, make all the related assumptions that you may need.

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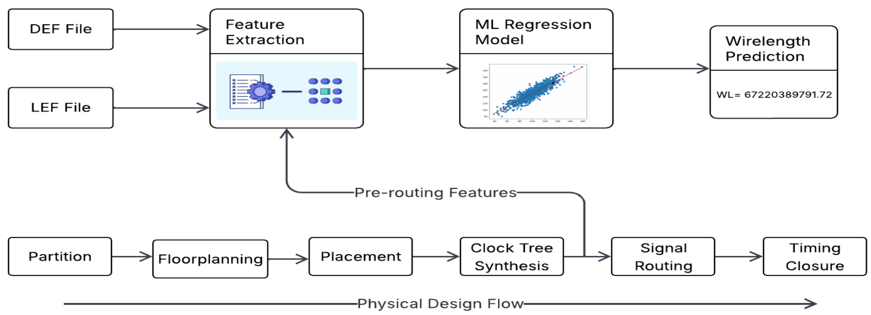
1. A methodology for early-stage wirelength prediction during the PD flow is shown below:



   (a) List out the features that would be needed for the above technique. (b) Through theoretical analysis, rank the above features for their suitability? Why/Why is this ranking not related to the ML method that you are choosing? How/How not? [3 + 3 + 2]

2. A key claim of PowerNet architecture (referring to the paper "PowerNet: Transferable Dynamic IR Drop Estimation via Maximum Convolutional Neural Network") is design independence (transferability), which most prior ML methods lack. (a) Explain what "design independence" means in the context of IR drop estimation. (b) The paper argues that directly using cell coordinates and timing as model features causes overfitting to training designs. Why? What does PowerNet do differently to avoid this problem? (c) In such IR-drop experiments, how are the training and test sets partitioned to ensure a fair evaluation of transferability? [3+3+3]

3. During the physical design stage of an advanced-node VLSI chip, a large amount of historical project data is available, including routed layout clips, placement density maps, congestion maps, timing-critical regions, power grid information, standard-cell distribution, netlist connectivity features, and final DRC violation reports from signoff tools. The design team wants to use this data to train a machine-learning model that can predict potential DRC hotspot regions before final signoff, so that layout engineers can fix risky areas earlier and reduce the number of expensive design-closure iterations.

   Explain how you would design a learning-based DRC hotspot prediction system for this scenario. In your answer, discuss what input features should be extracted from the physical design database, how DRC hotspot labels can be generated from historical signoff reports, what type of machine-learning or deep-learning model would be suitable, how the training/validation/testing data should be prepared, and how prediction accuracy should be evaluated. Also explain how such a model can handle probable limitations, such as data imbalance, design-to-design variation, false positives, and technology-node dependency. [3 + 2 + 2 + 4]

4. A semiconductor company is developing a multicore AI accelerator SoC for edge inference. The chip contains a RISC-V control processor, a DMA engine, an AXI-based interconnect, four neural processing cores, shared SRAM banks, a DDR memory controller, and several low-power clock-gated regions. During pre-silicon verification, the team uses a System Verilog(SV)-based

verification environment with constrained-random tests, protocol checkers, assertions, functional coverage models, and nightly regression runs.

After several RTL releases, the verification team observes that most normal traffic tests pass, but bugs are still being discovered late in scenarios involving simultaneous DMA transfers, cache-coherent memory access, AXI burst transactions, clock-domain crossings, and low-power mode entry/exit. Some failures appear only after long simulations and are difficult to reproduce. The project database contains historical RTL versions, UVM test configurations, random seeds, simulation logs, assertion failures, waveform traces, functional coverage reports, bug-tracking records, code commits, and previous regression results. The company wants to build a deep learning-based early verification closure assistant that can predict which blocks and test scenarios are most likely to contain unresolved bugs before full regression completion.

Propose suitable deep learning models for the following tasks:

- (a) Predicting whether a newly modified RTL block is high-risk.
- (b) Predicting which AXI/DMA traffic scenarios are likely to fail.
- (c) Detecting anomalous waveform behavior in long simulations.
- (d) Predicting uncovered or rarely covered functional scenarios.
- (e) Recommending new constrained-random or directed tests.

Since RTL changes weekly, explain how the model should deal with evolving RTL versions, new commits, unseen design blocks, and changing coverage goals? [10+ 3]

5. A semiconductor company is designing a chiplet-based AI inference SoC for edge computing. The SoC contains a RISC-V control subsystem, four neural processing units, an SRAM macro cluster, a DMA engine, a NoC router, PCIe interface, DDR controller, clock-generation unit, and power-management unit. The design target is high performance under strict area, timing, thermal, and power constraints. During early physical design, the team observes that manual floorplanning takes several iterations because the placement of SRAM macros, NoC routers, compute cores, and I/O interfaces strongly affects wirelength, congestion, timing closure, IR-drop risk, and thermal hotspots. Historical project data is available from earlier SoC versions, including macro placements, block aspect ratios, connectivity graphs, timing reports, congestion maps, power maps, thermal maps, routing overflow reports, and final floorplan quality metrics.

   The company wants to build a deep learning-based floorplanning assistant that can suggest an initial floorplan before detailed placement and routing.

   - (a) Propose a suitable deep learning architecture for the above scenario with detailed justification. For example, discuss whether graph neural networks, CNN-based layout encoders, transformers, reinforcement learning, or hybrid models are appropriate for learning macro placement and floorplan quality prediction.
   - (b) Explain how training labels or reward signals can be generated from historical floorplans. Include metrics such as total wirelength, congestion score, timing slack, IR-drop risk, thermal hotspot severity, routing overflow, and area utilization.
   - (c) The AI accelerator contains four neural processing units that communicate heavily with the SRAM macro cluster and NoC router. Explain how the model should learn to place these blocks to reduce communication latency and congestion. [3 + 3 + 3]